Data211-FinalExam

Neetu Shrestha Pradhan

2026-04-23

Loading the libraries.

```
library(readxl)

## Warning: package 'readxl' was built under R version 4.4.3

library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.4.3

library(rvest)

## Warning: package 'rvest' was built under R version 4.4.3

library(tidyverse)

## ── Attaching core tidyverse packages ──────────────────────── tidyverse
2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.5
## ✓ forcats   1.0.0     ✓ stringr   1.5.1
## ✓ lubridate 1.9.4     ✓ tibble    3.2.1
## ✓ purrr     1.0.2     ✓ tidyr     1.3.1
## ── Conflicts ────────────────────────────────────────
tidyverse_conflicts() ──
## ✗ dplyr::filter()         masks stats::filter()
## ✗ readr::guess_encoding() masks rvest::guess_encoding()
## ✗ dplyr::lag()            masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(readr)
library(scales)

## Warning: package 'scales' was built under R version 4.4.3

##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
```

```
##
##     col_factor
```

```r
library(Lahman)
```

```
## Warning: package 'Lahman' was built under R version 4.4.3
```

Loading the datasets.

```r
Learning<- read_excel("C:/Users/amark/OneDrive - MNSCU/data211 spring 2026/Final Exam File/Learning.xlsx")
MathScience_Scores<-read_excel("C:/Users/amark/OneDrive - MNSCU/data211 spring 2026/Final Exam File/MathScienceScores (1).xlsx")
Study_Habits<-read_excel("C:/Users/amark/OneDrive - MNSCU/data211 spring 2026/Final Exam File/StudyHabits.xlsx")
Tire_data<-read_excel("C:/Users/amark/OneDrive - MNSCU/data211 spring 2026/Final Exam File/TireData.XLSX")
```

1) creating the student_score data set

#```{r} #a #set.seed(123) #student_scores<- data.frame(#ID= 1:30,Score= sample(40:100,30,replace =TRUE)

1) #student_scores #b #head(student_scores, 10)

#c #student_scores_category<- data.frame(Score= student_scores$Score) #category = character(length =student_scores$Score) #student_scores$category<-c () # #for(i in 1:length(student_scores$Score)){ #if student_scores$Score[i] >= 85){

#student_scores$category[i] <- "Excellent" #else if student_scores$Score[i]>= 70{ #student_scores$category[i]<- “Good”

#else if student_scores$score[i] >= 50{ #student_scores$category[i]<- “Average”

#else student_scores$score[i] < 50student_scores$category[i] <-“Fail”{ # }

head(student_scores_category,10) #e) student_scores_category%>% group_by(category)%>% summaries(count=n())

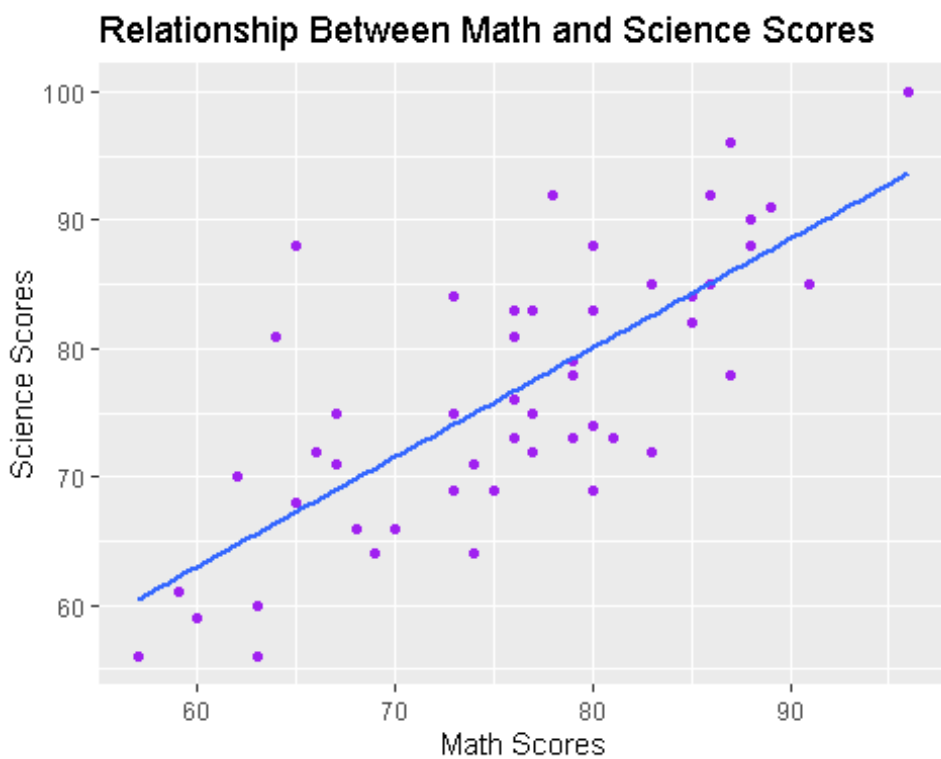
```
## question2)

```` r
#a)
MathScience_Scores<-read_excel("C:/Users/amark/OneDrive - MNSCU/data211 spring 2026/Final Exam File/MathScienceScores (1).xlsx")
sapply(MathScience_Scores[,c("Math","Science")],mean,na.rm = TRUE)
```

```
##    Math Science
##   75.84   76.50
```
```

```r
#b
ggplot (data = MathScience_Scores, aes(x = Math, y= Science))+
  geom_point(color ="purple")+
labs (x ="Math Scores",y ="Science Scores",
  title = "Relationship Between Math and Science Scores")+
  geom_smooth(method= "lm",se = FALSE)

## `geom_smooth()` using formula = 'y ~ x'
```




```r
theme_minimal()

## <theme> List of 144
##  $ line                            : <ggplot2::element_line>
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ lineend      : chr "butt"
##   ..@ linejoin     : chr "round"
##   ..@ arrow        : logi FALSE
##   ..@ arrow.fill   : chr "black"
##   ..@ inherit.blank: logi TRUE
##  $ rect                            : <ggplot2::element_rect>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
```

```
##   ..@ inherit.blank: logi TRUE
##  $ text                            : <ggplot2::element_text>
##   ..@ family       : chr ""
##   ..@ face         : chr "plain"
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : chr "black"
##   ..@ size         : num 11
##   ..@ hjust        : num 0.5
##   ..@ vjust        : num 0.5
##   ..@ angle        : num 0
##   ..@ lineheight   : num 0.9
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 0
##   ..@ debug        : logi FALSE
##   ..@ inherit.blank: logi TRUE
##  $ title                           : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ point                           : <ggplot2::element_point>
##   ..@ colour       : chr "black"
##   ..@ shape        : num 19
##   ..@ size         : num 1.5
##   ..@ fill         : chr "white"
##   ..@ stroke       : num 0.5
##   ..@ inherit.blank: logi TRUE
##  $ polygon                         : <ggplot2::element_polygon>
##   ..@ fill         : chr "white"
##   ..@ colour       : chr "black"
##   ..@ linewidth    : num 0.5
##   ..@ linetype     : num 1
##   ..@ linejoin     : chr "round"
##   ..@ inherit.blank: logi TRUE
##  $ geom                            : <ggplot2::element_geom>
##   ..@ ink        : chr "black"
##   ..@ paper      : chr "white"
##   ..@ accent     : chr "#3366FF"
##   ..@ linewidth  : num 0.5
```

```
##   ..@ borderwidth: num 0.5
##   ..@ linetype   : int 1
##   ..@ bordertype : int 1
##   ..@ family     : chr ""
##   ..@ fontsize   : num 3.87
##   ..@ pointsize  : num 1.5
##   ..@ pointshape : num 19
##   ..@ colour     : NULL
##   ..@ fill       : NULL
##  $ spacing                         : 'simpleUnit' num 5.5points
##   ..- attr(*, "unit")= int 8
##  $ margins                         : <ggplot2::margin> num [1:4] 5.5 5.5
5.5 5.5
##  $ aspect.ratio                    : NULL
##  $ axis.title                      : NULL
##  $ axis.title.x                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.75 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.top                : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 0
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.75 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.x.bottom             : NULL
##  $ axis.title.y                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
```

```
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num 90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.75 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.title.y.left               : NULL
##  $ axis.title.y.right              : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : num -90
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.75
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text                       : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : chr "#4D4D4DFF"
##   ..@ size         : 'rel' num 0.8
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
```

```
##   ..@ hjust        : NULL
##   ..@ vjust        : num 1
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 2.2 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.top                : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 4.95 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.x.bottom             : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 4.95 0 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 1
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 0
```

```
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y.left                : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 4.95 0 0
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.y.right               : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 4.95
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.text.theta                 : NULL
##  $ axis.text.r                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0.5
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ axis.ticks                      : <ggplot2::element_blank>
##  $ axis.ticks.x                    : NULL
```

```
##  $ axis.ticks.x.top                : NULL
##  $ axis.ticks.x.bottom             : NULL
##  $ axis.ticks.y                    : NULL
##  $ axis.ticks.y.left               : NULL
##  $ axis.ticks.y.right              : NULL
##  $ axis.ticks.theta                : NULL
##  $ axis.ticks.r                    : NULL
##  $ axis.minor.ticks.x.top          : NULL
##  $ axis.minor.ticks.x.bottom       : NULL
##  $ axis.minor.ticks.y.left         : NULL
##  $ axis.minor.ticks.y.right        : NULL
##  $ axis.minor.ticks.theta          : NULL
##  $ axis.minor.ticks.r              : NULL
##  $ axis.ticks.length               : 'rel' num 0.5
##  $ axis.ticks.length.x             : NULL
##  $ axis.ticks.length.x.top         : NULL
##  $ axis.ticks.length.x.bottom      : NULL
##  $ axis.ticks.length.y             : NULL
##  $ axis.ticks.length.y.left        : NULL
##  $ axis.ticks.length.y.right       : NULL
##  $ axis.ticks.length.theta         : NULL
##  $ axis.ticks.length.r             : NULL
##  $ axis.minor.ticks.length         : 'rel' num 0.75
##  $ axis.minor.ticks.length.x       : NULL
##  $ axis.minor.ticks.length.x.top   : NULL
##  $ axis.minor.ticks.length.x.bottom: NULL
##  $ axis.minor.ticks.length.y       : NULL
##  $ axis.minor.ticks.length.y.left  : NULL
##  $ axis.minor.ticks.length.y.right : NULL
##  $ axis.minor.ticks.length.theta   : NULL
##  $ axis.minor.ticks.length.r       : NULL
##  $ axis.line                       : <ggplot2::element_blank>
##  $ axis.line.x                     : NULL
##  $ axis.line.x.top                 : NULL
##  $ axis.line.x.bottom              : NULL
##  $ axis.line.y                     : NULL
##  $ axis.line.y.left                : NULL
##  $ axis.line.y.right               : NULL
##  $ axis.line.theta                 : NULL
##  $ axis.line.r                     : NULL
##  $ legend.background               : <ggplot2::element_blank>
##  $ legend.margin                   : NULL
##  $ legend.spacing                  : 'rel' num 2
##  $ legend.spacing.x                : NULL
##  $ legend.spacing.y                : NULL
##  $ legend.key                      : <ggplot2::element_blank>
##  $ legend.key.size                 : 'simpleUnit' num 1.2lines
##   ..- attr(*, "unit")= int 3
##  $ legend.key.height               : NULL
##  $ legend.key.width                : NULL
```

```
##  $ legend.key.spacing              : NULL
##  $ legend.key.spacing.x            : NULL
##  $ legend.key.spacing.y            : NULL
##  $ legend.key.justification        : NULL
##  $ legend.frame                    : NULL
##  $ legend.ticks                    : NULL
##  $ legend.ticks.length             : 'rel' num 0.2
##  $ legend.axis.line                : NULL
##  $ legend.text                     : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : 'rel' num 0.8
##   ..@ hjust        : NULL
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.text.position            : NULL
##  $ legend.title                    : <ggplot2::element_text>
##   ..@ family       : NULL
##   ..@ face         : NULL
##   ..@ italic       : chr NA
##   ..@ fontweight   : num NA
##   ..@ fontwidth    : num NA
##   ..@ colour       : NULL
##   ..@ size         : NULL
##   ..@ hjust        : num 0
##   ..@ vjust        : NULL
##   ..@ angle        : NULL
##   ..@ lineheight   : NULL
##   ..@ margin       : NULL
##   ..@ debug        : NULL
##   ..@ inherit.blank: logi TRUE
##  $ legend.title.position           : NULL
##  $ legend.position                 : chr "right"
##  $ legend.position.inside          : NULL
##  $ legend.direction                : NULL
##  $ legend.byrow                    : NULL
##  $ legend.justification            : chr "center"
##  $ legend.justification.top        : NULL
##  $ legend.justification.bottom     : NULL
##  $ legend.justification.left       : NULL
##  $ legend.justification.right      : NULL
##  $ legend.justification.inside     : NULL
```

```
##   [list output truncated]
##  @ complete: logi TRUE
##  @ validate: logi TRUE
```

Question 30

p = the proprotion of students who is greater than 40% $H_0: p =$

```
table(Study_Habits$Cramming)

##
##  No Yes
##  41  59

prop.test(x= 59, n = 100, alternative = "greater", correct = FALSE)

##
##  1-sample proportions test without continuity correction
##
## data:  59 out of 100, null probability 0.5
## X-squared = 3.24, df = 1, p-value = 0.03593
## alternative hypothesis: true p is greater than 0.5
## 95 percent confidence interval:
##  0.5077672 1.0000000
## sample estimates:
##    p
## 0.59
```

In above case, the p_value is less(0.03593) than significant value (alpha=0.05), so we failed to reject the null hypothesis.

Question 4)

p1= the average right tire pressure p2 = the average left tire pressure $H_0: $p1= p2 vs $p1 > p2$

```
Tire_data<-read_excel("C:/Users/amark/OneDrive - MNSCU/data211 spring
2026/Final Exam File/TireData.XLSX")
New_Tire<-Tire_data%>%
  pivot_longer(cols=c(RightTire, LeftTire),
               names_to ="Side" ,
               values_to = "Pressure" )

New_Tire

## # A tibble: 36 × 2
##    Side      Pressure
##    <chr>        <dbl>
##  1 RightTire       48
##  2 LeftTire        42
##  3 RightTire       80
```

```
##  4 LeftTire        75
##  5 RightTire       34
##  6 LeftTire        24
##  7 RightTire       63
##  8 LeftTire        56
##  9 RightTire       51
## 10 LeftTire        52
## # ℹ 26 more rows
```

```r
#b
head(New_Tire,6)
```

```
## # A tibble: 6 × 2
##   Side      Pressure
##   <chr>        <dbl>
## 1 RightTire       48
## 2 LeftTire        42
## 3 RightTire       80
## 4 LeftTire        75
## 5 RightTire       34
## 6 LeftTire        24
```

```r
t.test(Tire_data$RightTire,Tire_data$LeftTire,alternative = "greater",paired=
TRUE)
```

```
##
##  Paired t-test
##
## data:  Tire_data$RightTire and Tire_data$LeftTire
## t = 2.1744, df = 17, p-value = 0.02204
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  0.4999096       Inf
## sample estimates:
## mean difference
##             2.5
```

The above data result show p-value = 0.02204, which is lower than the significant value(alpha =0.05)

Question 5

p1 = the mean petal.length of virginica p2 = the mean petal.length of versicolor $H_0: $p1= p2 vs $p1 > p2$

```r
iris
```

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
## 1            5.1         3.5          1.4         0.2     setosa
## 2            4.9         3.0          1.4         0.2     setosa
## 3            4.7         3.2          1.3         0.2     setosa
```

```
## 4            4.6         3.1          1.5         0.2     setosa
## 5            5.0         3.6          1.4         0.2     setosa
## 6            5.4         3.9          1.7         0.4     setosa
## 7            4.6         3.4          1.4         0.3     setosa
## 8            5.0         3.4          1.5         0.2     setosa
## 9            4.4         2.9          1.4         0.2     setosa
## 10           4.9         3.1          1.5         0.1     setosa
## 11           5.4         3.7          1.5         0.2     setosa
## 12           4.8         3.4          1.6         0.2     setosa
## 13           4.8         3.0          1.4         0.1     setosa
## 14           4.3         3.0          1.1         0.1     setosa
## 15           5.8         4.0          1.2         0.2     setosa
## 16           5.7         4.4          1.5         0.4     setosa
## 17           5.4         3.9          1.3         0.4     setosa
## 18           5.1         3.5          1.4         0.3     setosa
## 19           5.7         3.8          1.7         0.3     setosa
## 20           5.1         3.8          1.5         0.3     setosa
## 21           5.4         3.4          1.7         0.2     setosa
## 22           5.1         3.7          1.5         0.4     setosa
## 23           4.6         3.6          1.0         0.2     setosa
## 24           5.1         3.3          1.7         0.5     setosa
## 25           4.8         3.4          1.9         0.2     setosa
## 26           5.0         3.0          1.6         0.2     setosa
## 27           5.0         3.4          1.6         0.4     setosa
## 28           5.2         3.5          1.5         0.2     setosa
## 29           5.2         3.4          1.4         0.2     setosa
## 30           4.7         3.2          1.6         0.2     setosa
## 31           4.8         3.1          1.6         0.2     setosa
## 32           5.4         3.4          1.5         0.4     setosa
## 33           5.2         4.1          1.5         0.1     setosa
## 34           5.5         4.2          1.4         0.2     setosa
## 35           4.9         3.1          1.5         0.2     setosa
## 36           5.0         3.2          1.2         0.2     setosa
## 37           5.5         3.5          1.3         0.2     setosa
## 38           4.9         3.6          1.4         0.1     setosa
## 39           4.4         3.0          1.3         0.2     setosa
## 40           5.1         3.4          1.5         0.2     setosa
## 41           5.0         3.5          1.3         0.3     setosa
## 42           4.5         2.3          1.3         0.3     setosa
## 43           4.4         3.2          1.3         0.2     setosa
## 44           5.0         3.5          1.6         0.6     setosa
## 45           5.1         3.8          1.9         0.4     setosa
## 46           4.8         3.0          1.4         0.3     setosa
## 47           5.1         3.8          1.6         0.2     setosa
## 48           4.6         3.2          1.4         0.2     setosa
## 49           5.3         3.7          1.5         0.2     setosa
## 50           5.0         3.3          1.4         0.2     setosa
## 51           7.0         3.2          4.7         1.4 versicolor
## 52           6.4         3.2          4.5         1.5 versicolor
## 53           6.9         3.1          4.9         1.5 versicolor
```

```
## 54            5.5          2.3           4.0          1.3 versicolor
## 55            6.5          2.8           4.6          1.5 versicolor
## 56            5.7          2.8           4.5          1.3 versicolor
## 57            6.3          3.3           4.7          1.6 versicolor
## 58            4.9          2.4           3.3          1.0 versicolor
## 59            6.6          2.9           4.6          1.3 versicolor
## 60            5.2          2.7           3.9          1.4 versicolor
## 61            5.0          2.0           3.5          1.0 versicolor
## 62            5.9          3.0           4.2          1.5 versicolor
## 63            6.0          2.2           4.0          1.0 versicolor
## 64            6.1          2.9           4.7          1.4 versicolor
## 65            5.6          2.9           3.6          1.3 versicolor
## 66            6.7          3.1           4.4          1.4 versicolor
## 67            5.6          3.0           4.5          1.5 versicolor
## 68            5.8          2.7           4.1          1.0 versicolor
## 69            6.2          2.2           4.5          1.5 versicolor
## 70            5.6          2.5           3.9          1.1 versicolor
## 71            5.9          3.2           4.8          1.8 versicolor
## 72            6.1          2.8           4.0          1.3 versicolor
## 73            6.3          2.5           4.9          1.5 versicolor
## 74            6.1          2.8           4.7          1.2 versicolor
## 75            6.4          2.9           4.3          1.3 versicolor
## 76            6.6          3.0           4.4          1.4 versicolor
## 77            6.8          2.8           4.8          1.4 versicolor
## 78            6.7          3.0           5.0          1.7 versicolor
## 79            6.0          2.9           4.5          1.5 versicolor
## 80            5.7          2.6           3.5          1.0 versicolor
## 81            5.5          2.4           3.8          1.1 versicolor
## 82            5.5          2.4           3.7          1.0 versicolor
## 83            5.8          2.7           3.9          1.2 versicolor
## 84            6.0          2.7           5.1          1.6 versicolor
## 85            5.4          3.0           4.5          1.5 versicolor
## 86            6.0          3.4           4.5          1.6 versicolor
## 87            6.7          3.1           4.7          1.5 versicolor
## 88            6.3          2.3           4.4          1.3 versicolor
## 89            5.6          3.0           4.1          1.3 versicolor
## 90            5.5          2.5           4.0          1.3 versicolor
## 91            5.5          2.6           4.4          1.2 versicolor
## 92            6.1          3.0           4.6          1.4 versicolor
## 93            5.8          2.6           4.0          1.2 versicolor
## 94            5.0          2.3           3.3          1.0 versicolor
## 95            5.6          2.7           4.2          1.3 versicolor
## 96            5.7          3.0           4.2          1.2 versicolor
## 97            5.7          2.9           4.2          1.3 versicolor
## 98            6.2          2.9           4.3          1.3 versicolor
## 99            5.1          2.5           3.0          1.1 versicolor
## 100           5.7          2.8           4.1          1.3 versicolor
## 101           6.3          3.3           6.0          2.5  virginica
## 102           5.8          2.7           5.1          1.9  virginica
## 103           7.1          3.0           5.9          2.1  virginica
```

```
## 104          6.3          2.9          5.6          1.8  virginica
## 105          6.5          3.0          5.8          2.2  virginica
## 106          7.6          3.0          6.6          2.1  virginica
## 107          4.9          2.5          4.5          1.7  virginica
## 108          7.3          2.9          6.3          1.8  virginica
## 109          6.7          2.5          5.8          1.8  virginica
## 110          7.2          3.6          6.1          2.5  virginica
## 111          6.5          3.2          5.1          2.0  virginica
## 112          6.4          2.7          5.3          1.9  virginica
## 113          6.8          3.0          5.5          2.1  virginica
## 114          5.7          2.5          5.0          2.0  virginica
## 115          5.8          2.8          5.1          2.4  virginica
## 116          6.4          3.2          5.3          2.3  virginica
## 117          6.5          3.0          5.5          1.8  virginica
## 118          7.7          3.8          6.7          2.2  virginica
## 119          7.7          2.6          6.9          2.3  virginica
## 120          6.0          2.2          5.0          1.5  virginica
## 121          6.9          3.2          5.7          2.3  virginica
## 122          5.6          2.8          4.9          2.0  virginica
## 123          7.7          2.8          6.7          2.0  virginica
## 124          6.3          2.7          4.9          1.8  virginica
## 125          6.7          3.3          5.7          2.1  virginica
## 126          7.2          3.2          6.0          1.8  virginica
## 127          6.2          2.8          4.8          1.8  virginica
## 128          6.1          3.0          4.9          1.8  virginica
## 129          6.4          2.8          5.6          2.1  virginica
## 130          7.2          3.0          5.8          1.6  virginica
## 131          7.4          2.8          6.1          1.9  virginica
## 132          7.9          3.8          6.4          2.0  virginica
## 133          6.4          2.8          5.6          2.2  virginica
## 134          6.3          2.8          5.1          1.5  virginica
## 135          6.1          2.6          5.6          1.4  virginica
## 136          7.7          3.0          6.1          2.3  virginica
## 137          6.3          3.4          5.6          2.4  virginica
## 138          6.4          3.1          5.5          1.8  virginica
## 139          6.0          3.0          4.8          1.8  virginica
## 140          6.9          3.1          5.4          2.1  virginica
## 141          6.7          3.1          5.6          2.4  virginica
## 142          6.9          3.1          5.1          2.3  virginica
## 143          5.8          2.7          5.1          1.9  virginica
## 144          6.8          3.2          5.9          2.3  virginica
## 145          6.7          3.3          5.7          2.5  virginica
## 146          6.7          3.0          5.2          2.3  virginica
## 147          6.3          2.5          5.0          1.9  virginica
## 148          6.5          3.0          5.2          2.0  virginica
## 149          6.2          3.4          5.4          2.3  virginica
## 150          5.9          3.0          5.1          1.8  virginica
```

```r
iris2 <- iris%>%
 filter(Species == c("versicolor", "virginica"))
iris2
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
## 1           7.0         3.2          4.7         1.4 versicolor
## 2           6.9         3.1          4.9         1.5 versicolor
## 3           6.5         2.8          4.6         1.5 versicolor
## 4           6.3         3.3          4.7         1.6 versicolor
## 5           6.6         2.9          4.6         1.3 versicolor
## 6           5.0         2.0          3.5         1.0 versicolor
## 7           6.0         2.2          4.0         1.0 versicolor
## 8           5.6         2.9          3.6         1.3 versicolor
## 9           5.6         3.0          4.5         1.5 versicolor
## 10          6.2         2.2          4.5         1.5 versicolor
## 11          5.9         3.2          4.8         1.8 versicolor
## 12          6.3         2.5          4.9         1.5 versicolor
## 13          6.4         2.9          4.3         1.3 versicolor
## 14          6.8         2.8          4.8         1.4 versicolor
## 15          6.0         2.9          4.5         1.5 versicolor
## 16          5.5         2.4          3.8         1.1 versicolor
## 17          5.8         2.7          3.9         1.2 versicolor
## 18          5.4         3.0          4.5         1.5 versicolor
## 19          6.7         3.1          4.7         1.5 versicolor
## 20          5.6         3.0          4.1         1.3 versicolor
## 21          5.5         2.6          4.4         1.2 versicolor
## 22          5.8         2.6          4.0         1.2 versicolor
## 23          5.6         2.7          4.2         1.3 versicolor
## 24          5.7         2.9          4.2         1.3 versicolor
## 25          5.1         2.5          3.0         1.1 versicolor
## 26          5.8         2.7          5.1         1.9  virginica
## 27          6.3         2.9          5.6         1.8  virginica
## 28          7.6         3.0          6.6         2.1  virginica
## 29          7.3         2.9          6.3         1.8  virginica
## 30          7.2         3.6          6.1         2.5  virginica
## 31          6.4         2.7          5.3         1.9  virginica
## 32          5.7         2.5          5.0         2.0  virginica
## 33          6.4         3.2          5.3         2.3  virginica
## 34          7.7         3.8          6.7         2.2  virginica
## 35          6.0         2.2          5.0         1.5  virginica
## 36          5.6         2.8          4.9         2.0  virginica
## 37          6.3         2.7          4.9         1.8  virginica
## 38          7.2         3.2          6.0         1.8  virginica
## 39          6.1         3.0          4.9         1.8  virginica
## 40          7.2         3.0          5.8         1.6  virginica
## 41          7.9         3.8          6.4         2.0  virginica
## 42          6.3         2.8          5.1         1.5  virginica
## 43          7.7         3.0          6.1         2.3  virginica
## 44          6.4         3.1          5.5         1.8  virginica
## 45          6.9         3.1          5.4         2.1  virginica
```

```
## 46          6.9         3.1          5.1         2.3  virginica
## 47          6.8         3.2          5.9         2.3  virginica
## 48          6.7         3.0          5.2         2.3  virginica
## 49          6.5         3.0          5.2         2.0  virginica
## 50          5.9         3.0          5.1         1.8  virginica
```

```
#b
iris2[60:70,]
```

```
##       Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## NA              NA          NA           NA          NA    <NA>
## NA.1            NA          NA           NA          NA    <NA>
## NA.2            NA          NA           NA          NA    <NA>
## NA.3            NA          NA           NA          NA    <NA>
## NA.4            NA          NA           NA          NA    <NA>
## NA.5            NA          NA           NA          NA    <NA>
## NA.6            NA          NA           NA          NA    <NA>
## NA.7            NA          NA           NA          NA    <NA>
## NA.8            NA          NA           NA          NA    <NA>
## NA.9            NA          NA           NA          NA    <NA>
## NA.10           NA          NA           NA          NA    <NA>
```

```
#prop.test(x=c(60,70), n = (130,130) alternative = "greater", correct =
FALSE)
```

Question 6

P1= proportion of students preference in-person class p2 = proportion of student preference online class $H_0: p1 = p2$ vs $p1 > p2$

```
table(Learning$Preference)
```

```
##
## InPerson   Online
##       87      133
```

```
Learning%>%
  group_by(Group)%>%
  summarise(total=n())
```

```
## # A tibble: 2 × 2
##   Group total
##   <chr> <int>
## 1 A       120
## 2 B       100
```

```
prop.test(x = c(120,100),n=c(220,220),alternative = "two.sided" ,correct =
FALSE)
```

```
##
##  2-sample test for equality of proportions without continuity correction
##
```

```
## data:  c(120, 100) out of c(220, 220)
## X-squared = 3.6364, df = 1, p-value = 0.05653
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.002141619  0.183959801
## sample estimates:
##    prop 1    prop 2
## 0.5454545 0.4545455
```

Learning

```
## # A tibble: 220 × 2
##    Group Preference
##    <chr> <chr>
##  1 A     Online
##  2 A     Online
##  3 A     Online
##  4 A     Online
##  5 A     Online
##  6 A     Online
##  7 A     Online
##  8 A     Online
##  9 A     Online
## 10 A     Online
## # ℹ 210 more rows
```

In this above case, the p_value is 0.05653 which is slidly higer than the significant value of alpha = 0.05. In this scenorio, we can say that it fail to reject the null hyphothsis.